Jonathan Zhu

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SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL
Backend / Systems: FastAPI, Node.js, Express.js, REST APIs, JWT, OAuth2, Pydantic, SQLAlchemy
Frontend: React, Next.js, TypeScript, Tailwind CSS, Material UI, React Query, Framer Motion
Databases: PostgreSQL, MySQL, MongoDB
Cloud / Infrastructure: AWS (EC2, RDS, S3, Bedrock, Cognito, API Gateway, Amplify, Lambda, Secrets Manager, VPC), Terraform
AI / ML: Amazon Nova Pro, Anthropic API, OpenAI API, Gemini API, Prompt Engineering, Multimodal LLMs
Tools & Compliance: Jenkins, Splunk, Dynatrace, HIPAA Compliance, BAA, pdf2image, asyncpg

WORK EXPERIENCE

Citywide Eye Care Great Neck, NY
Full-Stack Software Engineer *Jan 2024 – Present*

- Lead engineer for a production EMR platform used daily by clinical staff, architecting a zero-downtime migration from on-premise servers to AWS (EC2, RDS, S3, Cognito, API Gateway, Amplify)
- Owned HIPAA security architecture end to end: Cognito JWT auth, IP whitelisting, role-based access control, and WireGuard VPN, passing all external compliance reviews with zero audit findings
- Reduced physician documentation time by 75% by engineering structured charting workflows with auto-save, copy-forward, and S3-backed document storage via pre-signed URLs
- Partnered directly with clinicians to define requirements and iterate on features, accelerating feedback cycles and cutting rework across all major releases

ZLA (Client Contract – AI-Enabled CRM Platform) Rochester, NY
Full-Stack Software Engineer *Aug 2025 – Present*

- Built a production CRM from scratch on a 5-person Agile team, delivering lead management, property tracking, campaign workflows, and agent dashboards within tight contract timelines
- Engineered an AI-powered lead discovery pipeline integrating Google Places API, Geocoding, Brave Search, and OpenAI, cutting manual prospect research time by 60%
- Architected a normalized FastAPI and PostgreSQL backend with complex relational schemas; built a React/TypeScript frontend with Kanban pipelines and map-based lead search
- Provisioned and managed AWS infrastructure with Terraform (Amplify, EC2 with Elastic IP, API Gateway, Lambda) and built an admin dashboard for one-click backend server management

Citywide Eye Care Great Neck, NY
Software Engineer Intern *Sep 2023 – Dec 2023*

- Built and deployed an AI health chatbot using Python and Gemini API, autonomously handling patient FAQs and symptom triage, cutting front-desk call volume by 50%
- Rebuilt the patient-facing website in Next.js and Material UI, improving SEO and driving a 25% increase in appointment conversion rates
- Implemented digital intake and scheduling workflows on AWS (EC2, S3, RDS, Secrets Manager), replacing all manual paper-based processes

ADP Roseland, NJ
Application Developer Intern *Jun 2023 – Aug 2023*

- Built an automated status-reporting tool adopted across the team, eliminating manual report generation and cutting decision turnaround time by 20%
- Reduced deployment time by 30% by optimizing the Jenkins CI/CD pipeline and improving MySQL query performance across internal analytics integrations
- Enhanced S3, Splunk, and Dynatrace integrations supporting internal analytics; recognized by engineering leadership for measurable impact on team velocity

PROJECTS

- **RemitAI:** HIPAA-compliant AI platform for healthcare RCM using Amazon Nova Pro on AWS Bedrock (BAA); processes EOB/ERA PDFs via multimodal pipeline, classifies denials by CARC/RARC codes, generates appeal strategies, and builds a self-learning payer knowledge base on async FastAPI, PostgreSQL, and React/TypeScript
- **AI Lead Discovery Pipeline:** Multi-source enrichment pipeline integrating Google Places API, Geocoding, Brave Search, and OpenAI to surface qualified real estate prospects, cutting manual research time by 60%
- **Enlighten Learning:** Full-stack nonprofit site built in React, TypeScript, and Material UI with serverless API integration (Google Apps Script), dark/light mode, animated UI, and donation tracking; supports tutoring programs for students in underserved communities across Long Island

EDUCATION

Georgia Institute of Technology Admitted, Fall 2026
M.S. in Computer Science

Rochester Institute of Technology Expected May 2026
B.S. in Software Engineering